

AI Based PPE Detection System

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Introduction

Workplace safety is a critical concern across high-risk industries such as chemical processing, construction, manufacturing, and shipbuilding. These environments often involve hazardous operations where strict compliance with Personal Protective Equipment (PPE) regulations is essential to prevent serious injuries and fatalities.

However, current PPE monitoring practices are largely manual, making them inconsistent, labor-intensive, and highly susceptible to human error. In large-scale or fast-paced industrial settings, ensuring 100% compliance through manual supervision becomes impractical. Moreover, traditional safety checks typically lack real-time responsiveness, comprehensive data logging, and scalability.

To address these limitations, this project proposes the development of an AI Based PPE Detection System. The system is designed to automatically identify critical PPE components such as helmets, gloves, masks, and safety vests on workers in real time. It will also be capable of issuing instant alerts for non-compliance and generating actionable compliance reports to assist safety officers in enforcing safety standards efficiently.

This solution not only enhances real-time safety monitoring but also promotes a proactive safety culture by providing data-driven insights for continuous improvement. Given the increasing emphasis on smart industry practices and workplace safety, this project idea is both timely and highly relevant to modern industrial needs.

Why Was This Product Selected?

The AI-based PPE Detection System was selected to address critical safety gaps and inefficiencies in industrial and construction settings, particularly within the Bangladeshi industrial sector.

Addressing Safety Gaps and Human Error: Traditional manual monitoring of Personal Protective Equipment (PPE) compliance is often time-consuming, inconsistent, and highly prone to human error, making it practically impossible to maintain safety standards in large industrial environments. The lack of AI-based PPE detectors significantly impacts workplace safety and operational efficiency, leading to an increased risk of workplace accidents and injuries. There is a need to prevent workplace injuries and occupational hazards that require 24/7 monitoring capabilities beyond human capacity.

Need for Real-Time Monitoring: There is an uncovered demand for automated PPE monitoring, with 83.9% of surveyed participants feeling the need for an AI-based PPE detector. Real-time detection and immediate alerts are essential to minimize the time workers spend in dangerous situations without proper protection. An overwhelming majority (93.5%) of surveyed preferred having real-time alerts.

Industry-Specific Needs: Specific industries, such as chemical, pharmaceutical, and food processing, have unique PPE requirements that necessitate tailored detection capabilities.

Market Opportunity: The Bangladeshi industrial sectors, including construction, manufacturing, and logistics, are experiencing rapid growth, which increases the emphasis on workplace safety. This growth, coupled with a high level of awareness about basic PPE concepts among surveyed audiences (93.5% aware), indicates a fertile ground for an AI-based PPE detection system. The product aims to leverage object detection algorithm to accurately and instantly identify whether individuals are wearing essential protective gear, thereby strengthening safety enforcement and minimizing accident risks.

Feasibility of the Product

The feasibility of the AI-based PPE Detection System is supported by strong market demand, identified customer requirements, and a viable technical design and pricing strategy.

Market Awareness and Demand:

While 93.5% of surveyed participants were aware of PPE, a majority (67.7%) were not familiar with AI-based PPE detectors. However, a significant majority (83.9%) expressed a need for this technology.

A large majority of those surveyed (83.9%) recognize the importance of AI-based PPE detectors for modern industry, with 45.2% considering it "Important" and 38.7% "Very important".

The most frequently used PPE types are helmets (around 12 counts) and gloves (around 9 counts), while customers most frequently want to identify helmets (around 11 counts) and combinations including helmets, masks, and goggles.

Market Size and Growth:

1. The global PPE market is forecast to grow to USD 171.66 billion in 2034 at a CAGR of 7.24% from USD 91.51 billion in 2025, driven by rising safety awareness, industrial expansion, and regulatory policies.
2. In Bangladesh, the demand for PPE is growing due to industrial expansion, especially in construction, manufacturing, and healthcare.
3. The **Total Addressable Market (TAM)** for AI-powered PPE detection systems in Bangladesh is valued between **USD 1.2 to 1.5 billion**.
4. The **Serviceable Available Market (SAM)** is estimated at **USD 360 to 450 million**, covering industries with strict safety regulations like chemical, pharmaceutical, food & beverage, and shipbuilding.
5. The **Serviceable Obtainable Market (SOM)**, a realistic near-term market share, ranges from **USD 3.6 to 22.5 million**.

Customer Requirements and Prioritization (Kano Model & QFD):

1. The product design is driven by identified customer needs. Through the Kano Model, features were categorized based on their impact on customer satisfaction.
2. **Must-be Features:** Real-time alert, high accuracy, quick response when someone is not wearing PPE, functionality in low internet zones, portability, and works in both indoor and outdoor construction environments were identified as "must-be" features. These are essential to avoid dissatisfaction. For example, 93.5% of participants desire real-time alerts, and nearly all surveyed placed significant importance on the capability for multiple detections. High accuracy is considered important or very important by 83.4% of respondents.
3. **Attractive Features:** Multiple detection, multiple language support, battery or secondary power options, and integration with existing safety monitoring systems were classified as "attractive" features, meaning their presence greatly increases satisfaction.
4. **Performance Features:** Maintenance cost was identified as a "one-dimensional" (performance) feature, meaning more of it (e.g., lower cost) leads to more satisfaction.
5. **Durability and Range:** A majority (54.8%) considered long-term durability "Important". A majority (51.6%) also placed significant importance on the camera's detection range for

PPE. The system's ability to detect PPE violations remotely without requiring the worker to go to a specific spot is highly valued (61.3% "Important," 22.6% "Very Important").

6. **Reliability:** The reliability of the Kano model results was assessed using Cronbach's Alpha, which yielded a value of 0.617, falling into the "quite reliable" category, confirming the internal consistency of the survey

Pricing Strategy:

1. The **optimal price point** for the AI-based PPE Detection System is identified as **7000 BDT**.
2. The **acceptable price range** lies between **5500 BDT (Point of Marginal Cheapness)** and **7100 BDT (Point of Marginal Expensiveness)**.
3. The **indifference price point** is **6300 BDT**.
4. Ideally, the system should be priced between **6300 and 7000 BDT** to balance value perception, affordability, and customer acceptance.

Technical Feasibility (Component Selection):

1. The system uses readily available and suitable components like the **Raspberry Pi 3B+** as the main processor for running lightweight AI models, chosen for its performance, affordability, and ease of integration.
2. The **Raspberry Pi Camera Module v2** provides high-resolution imaging suitable for detecting safety gear.
3. Other components like MicroSD cards for storage, 5V buzzers and LEDs for alerts, resistors, GPIO header wires, a 10,000 mAh power bank for secondary power, and a custom enclosure ensure the system's functionality and durability. The inclusion of a power bank ensures uninterrupted operation during power cuts

How to Implement the Product

Implementation involves defining the functional structure, selecting specific components, establishing alert mechanisms, and considering deployment environments, all guided by customer requirements.

Functional Structure Development:

1. **Black Box:** The system is designed as a Black Box, taking visual input from a camera and producing output to a display monitor and alert units (LEDs, buzzer), with internal processing hidden from the user.
2. **Transparent Box:** Internally, the system processes camera input to detect PPE compliance and generates output signals for alerts and display.
3. **Function Tree:** The system captures visual input via a camera, pre-processes footage to detect human presence and PPE items, compares them against compliance criteria, makes a decision, and activates alerts (buzzer, LED) for non-compliance, while displaying compliance status in real-time. A power supply ensures all components function correctly.
4. **Function Structure:** The system is organized into sensing (Raspberry Pi Camera Module), processing (Raspberry Pi 3B+ running an ML model), and alerting blocks. A green LED indicates compliance, while a red LED and buzzer activate for non-compliance. All components are housed in a custom enclosure, powered by an AC adapter with a 10000mAh power bank backup.

Main Components and Specifications:

1. **Raspberry Pi 3B+:** 1.4GHz quad-core ARM Cortex-A53 processor, 1GB RAM, dual-band WiFi, Bluetooth 4.2, CSI camera interface, suitable for running lightweight AI models offline.
2. **Raspberry Pi Camera Module v2:** 8-megapixel Sony IMX219 sensor, capable of 3280×2464 stills and 1080p30 video, ideal for detecting safety gear with clarity.
3. **MicroSD Card (16–32 GB):** Primary storage for OS and AI models, ensuring reliable booting and smooth operation.
4. **5V Buzzer (Active):** Provides immediate audible alerts, easily controlled by Raspberry Pi's GPIO pins.

5. **Red LED (5mm) and Green LED (5mm):** Visual indicators for PPE violations (red) and compliance (green), offering clear, real-time feedback.
6. **220Ω Resistors:** Safely limit current to the LEDs, protecting components.
7. **GPIO Header Wires:** Enable flexible and secure connections between Raspberry Pi and external components.
8. **Power Bank (10,000 mAh):** Provides portable and reliable power, especially in areas with unstable sources.
9. **Micro-USB Cable:** Supplies stable power and enables data transfer for the Raspberry Pi.
10. **Mini Breadboard / PCB:** Facilitate prototyping and permanent circuit assembly.
11. **Case/Enclosure (Custom):** Houses and protects components, with features like ventilation and camera cutouts for rugged industrial environments.

Alert and Notification Systems:

1. Participants overwhelmingly prefer real-time alerts.
2. The majority (54.8%) prefer alerts via **Light (Red/Green)**, 29.0% favor **Sound**, and 16.1% opt for **SMS**. The system incorporates light (LEDs) and sound (buzzer) for alerts.

Installation and Deployment Considerations:

1. The majority of surveyed participants (51.6%) prefer the PPE detection camera to be placed on **walls**, followed by ceilings (32.3%) and ground (16.1%).
2. Most surveyed people highly value the system's ability to detect PPE violations **remotely** without requiring the worker to go to a specific spot.
3. The system needs to operate for **required hours** rather than continuously 24/7, as preferred by 77.4% of participants.
4. **Support for multiple languages** for user instructions, alerts, and interface is significantly valued by most surveyed (41% "Important," 29% "Very important").
5. **Offline processing capability** is crucial as many industrial facilities have limited internet connectivity.
6. **Integration with existing camera infrastructure** is preferred for cost-effectiveness and easier deployment.

Market Analysis

Potential Customers:

In today's fast-paced industrial environments, ensuring worker safety is more critical than ever. Traditional PPE monitoring methods often fall short in real-time enforcement, leading to potential risks and compliance issues. Our AI-based PPE detection system offers a smart, automated solution to monitor PPE usage across various sectors, minimizing human error and enhancing workplace safety. The table below highlights typical safety-critical areas and required PPE across major industries.

Market Analysis:

Bangladesh Market:

Bangladesh's industrial sectors—particularly construction, manufacturing, and logistics—are

Industry Sector	Typical Safety-Critical Areas	PPE Requirements
Pharmaceutical	Clean rooms, chemical labs	Masks, gloves, goggles, gowns
Food & Beverage	Processing & packaging lines	Gloves, masks, aprons
Construction	Building sites, scaffolding areas	Helmets, vests, boots, gloves
Textile & Garments	Dyeing units, chemical treatment areas	Gloves, masks, safety glasses
Manufacturing (General)	Assembly lines, machinery operation areas	Hard hats, gloves, ear protection
Logistics & Warehousing	Loading docks, cold storage, transport hubs	Safety vests, steel-toe boots, gloves, helmets

experiencing rapid growth. With this expansion comes an increased emphasis on workplace safety. [1] outlines the impact of lacking Personal Protective Equipment (PPE) on construction sites in Bangladesh,

including increased injuries, project delays, cost overruns, substandard work quality, negative effects on economic development, and reduced worker morale. It emphasizes the necessity of PPE for ensuring safety and efficient project execution.

The potential market for AI-powered PPE detection systems in Bangladesh can be categorized into three segments. The **Total Addressable Market (TAM)** is valued between USD 1.2 to 1.5 billion, representing the entire PPE market across various industrial sectors. The **Serviceable Available Market (SAM)**, estimated at USD 360 to 450 million, focuses on industries where PPE detection technologies are applicable, such as chemical, pharmaceutical, food & beverage, and shipbuilding. Lastly, **the Serviceable Obtainable Market (SOM)** ranges from USD 3.6 to 22.5 million, reflecting a realistic near-term market share based on current operational capacity, competition, and adoption trends. This segmentation specifically pertains to the AI-based PPE detection market in Bangladesh.

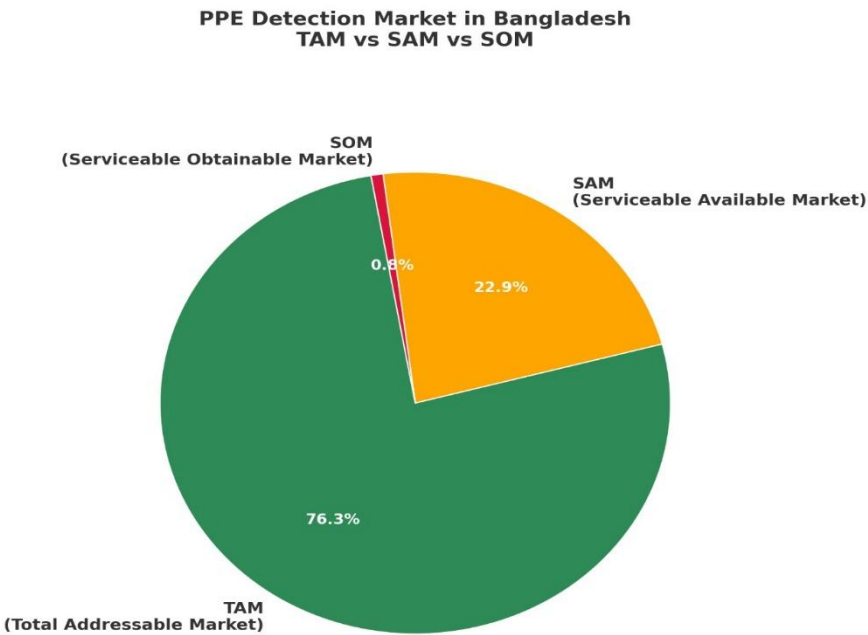


Figure 1 PPE Detection Market Segmentation in Bangladesh: TAM vs SAM vs SOM

Global Market:

The market for AI-based PPE detection systems is experiencing rapid growth, driven by increasing workplace safety demands, regulatory compliance, and advancements in computer vision technology.

According to a report by Market.us, the global market size for PPE detection was valued at approximately USD 311 million in 2023 and is projected to reach USD 117.4 billion by 2033, reflecting a staggering compound annual growth rate (CAGR) of around 81%. Similarly, a report by Straits Research forecasts growth from USD 1.80 billion in 2024 to USD 358.6 billion by 2033. This exponential rise underscores the massive potential of AI-driven safety systems and highlights the increasing reliance on intelligent automation to ensure real-time compliance and reduce workplace hazards.

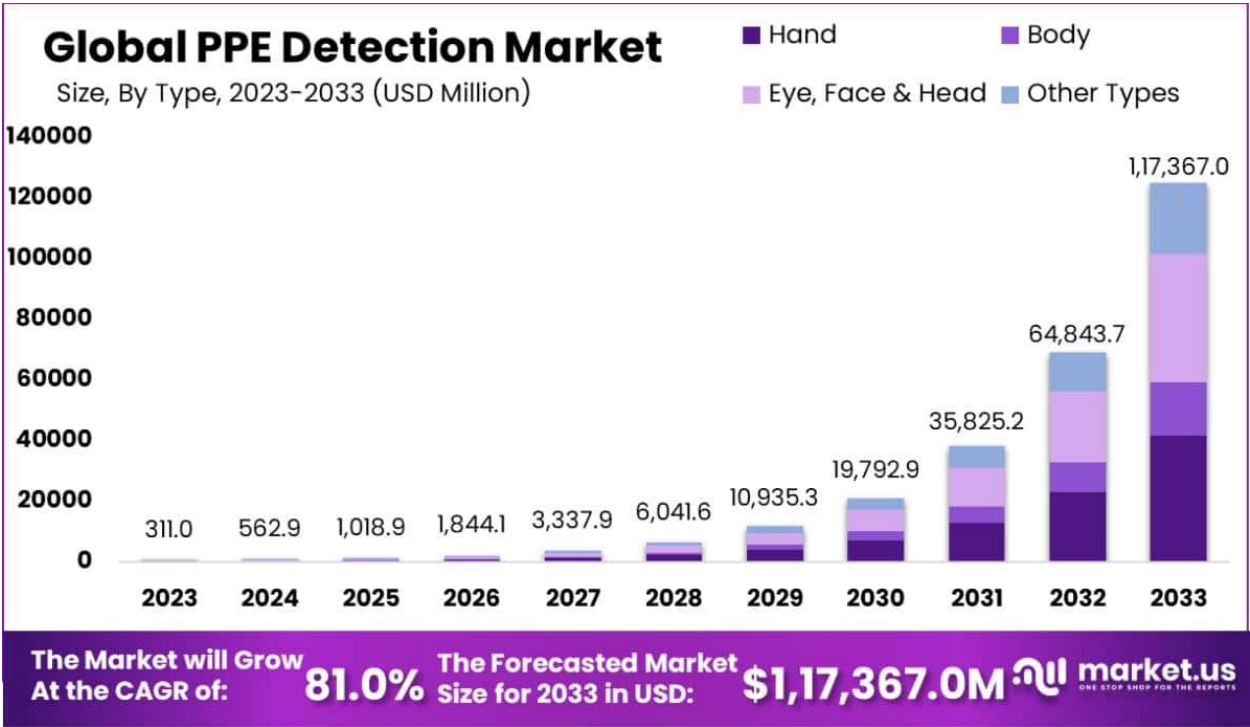


Figure 2 Projected Global PPE Detection Market Size by Segment, 2023–2033 (USD Million)

Target Audience:

Our primary target audience is the construction sector in Bangladesh, which accounts for a significant portion of workplace injuries and fatalities due to low compliance with PPE protocols. This sector includes construction firms, real estate developers, and public infrastructure contractors. AI-based PPE detection can play a transformative role in enhancing real-time safety monitoring and regulatory compliance.

Key Metric	Value / Insight
Number of Construction Workers	3.5 million+ (as of 2023)
Annual Growth Rate	6.5% (steady growth driven by urbanization and infrastructure projects)
Workplace Accident Rate	Highest among all industries (ILO & DIFE reports)
Estimated PPE Detection Market (Construction)	USD 1.2 – 1.5 million (SOM estimate)
Major Project Hubs	Dhaka, Chattogram, Rajshahi, Khulna, Sylhet
Common PPE Needed	Helmets, vests, gloves, boots

SWOT Analysis Table: PPE Detection Systems

Product	Strengths	Weaknesses	Opportunities	Threats
Vertex AI PPE Detection	1. Backed by DeepMind with cutting-edge AI research and resource 2. Built on Google Cloud’s scalable, enterprise-grade Vertex AI platform 3. Likely integrates well with existing clouds infrastructure and workflows.	1. Not tailored specifically for PPE -more of a general ML platform that requires significant customization . 2. Might lack real-time edge inference, leading to latency. 3. High costs.	1. Large Enterprise users can unify PPE detection with broader analytics pipelines. 2. Opportunity to add value via pertained models, AutoML and MLOps	1. Highly technical—requires in-house ML talent or consulting support. 2. Specialized competitors (e.g. Intenseye) offer ready-to-use PPE detection.

Protex	1. Self-serve and highly customizable—suits small to mid-market clients. 2. Modular approach allows tailoring to specific PPE items or workflows.	1. Complexity and heavy customization may hamper scalability in large enterprises. 2. Not industry-specific—potentially inconsistent accuracy across environments.	1. Could expand support with industry templates or marketplace models. 2. Offer vertical bundles e.g. for manufacturing.	1. Specialized turnkey systems outcompeting on ease-of-use and reliability. 2. Customization overhead may deter time-sensitive clients.
Intenseye	1. Purpose-built PPE compliance module, covering ~10 PPE types with high precision. 2. Quick deployment over existing CCTV (cloud, hybrid, on-prem) .	1. Likely subscription cost may be prohibitive for smaller operations. 2. Dependence on camera placement and visual conditions.	1. Expand into new industries (construction, logistics). 2. Add fine-grained detection (correct fit, sequence) .	1. Edge-processing challengers or localized inference competitors. 2. New solutions using vision-language models may

	3.. Integrates into EHS workflows with dashboards, scoring, automated corrective alerts. 4. Proven track record (e.g. 99% face-shield compliance increase).	3.High Price		offer richer context.
VisionPlatform	1.Offers free standard PPE models and full control via re-training. 2. Highly flexible: supports various camera types (RGB, thermal, lidar). 3. Edge/on-premise deployment ensures low latency and privacy.	1.Requires technical know-how—users must annotate data and retrain models. 2.Baseline model accuracy may be inconsistent in real-world conditions.	1.Can build a community for reusable PPE modules or datasets. 2.Offer managed services or consult to reduce deployment friction.	1.Easy-to-use commercial plug-and-play systems (e.g. Intenseye) may capture non-technical users. 2.Risk of low adoption if retraining burden is too high.
	1.Low price	1.New player, less	1.First mover	1.Intense competition

Our Product	2.Quick installation (Plug and play)	trust at first, limited funding	advantage at local market	of global competitor
	3.Can be used with existing service	2.Less-powerful hardware. 3.No cloud service support	2.Price to performance is attractive to customer	2.Regulatory challenges meeting diverse international standard

Design and Specification Analysis

Introduction:

Design and Specification Analysis is a critical phase in the product development and engineering design process. It involves evaluating the functionality, feasibility, performance, and cost-effectiveness of a proposed design against a set of defined requirements or specifications. This stage ensures that the final product not only meets customer needs and regulatory standards but also aligns with technical constraints, manufacturing capabilities, and economic considerations. Through systematic analysis of design alternatives and detailed specification assessment, this process aids in optimizing product quality, minimizing risks, and enhancing overall project success.

Specification Chart:

Table 1 Electrical Components

Parts Name	Description	Quantity
Raspberry Pi 3B+	Main processor	01
Raspberry Pi Camera Module v2	Image input	01
MicroSD Card (16–32 GB)	Storage for OS & model	01
5V Buzzer (Active)	Beep alert	01
Red LED (5mm)	PPE violation alert	01

Green LED (5mm)	PPE OK alert	01
220 Ω Resistors	For LED current limiting	02
GPIO Header Wires	For LED/Buzzer connection	~6
Power Bank (10,000 mAh)	Secondary power	01
Micro-USB Cable	Power connection	01
Mini Breadboard / PCB	For mounting components	01

Table 2 Structural Components

Case/Enclosure (Custom)	3D printed/assembled housing	01
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Main components of AI based PPE detection System:

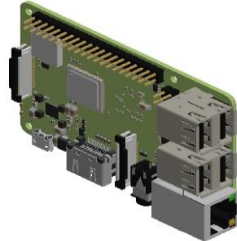
- 1) Raspberry Pi 3B+
- 2) Raspberry Pi Camera Module v2
- 3) MicroSD Card
- 4) 5V Buzzer
- 5) Green LED
- 6) Red LED
- 7) 220 Ω Resistors
- 8) GPIO Header Wires
- 9) Breadboard / PCB
- 10) Power Bank
- 11) Case/Enclosure

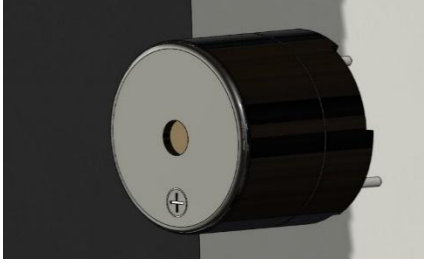
Component Dimensions:

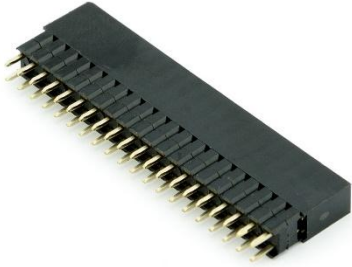
The following section provides detailed images of the key components used in the AI-based PPE detection system along with their precise physical dimensions. These visuals help in understanding the size, form factor, and integration requirements of each part, ensuring accurate design, assembly, and deployment in industrial environments. Clear knowledge of component dimensions is essential for creating effective mounts, enclosures, and ensuring seamless hardware compatibility. All the measurements are in millimeters.

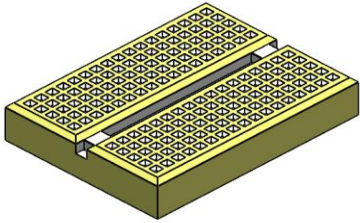
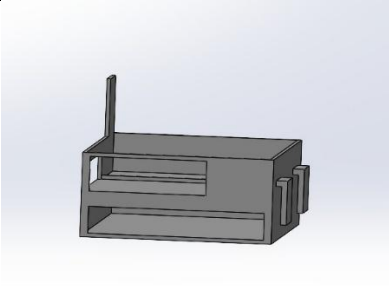
Summary of the Components

Table 3 Summary of the Components

Serial No	Component Name	Specifications	Isometric View
01	Raspberry Pi 3B+	CPU: 1.4 GHz Quad-Core ARM Cortex-A53 RAM: 1 GB LPDDR2 Wi-Fi: Dual-band 2.4 GHz & 5 GHz 802.11ac Ethernet: Gigabit (300 Mbps max) USB: 4 × USB 2.0 ports Storage: microSD slot Power: 5V / 2.5A (Micro-USB)	
02	Raspberry Pi Camera Module v2	Sensor: 8MP Sony IMX219 Image: 3280×2464 Video: 1080p30, 720p60, 480p90 Interface: CSI Lens: Fixed focus	
03	MicroSD Card (16–32 GB)	Storage: 16 GB to 32 GB Type: Class 10 / UHS-I Use: OS & file storage for Raspberry Pi Speed: Fast read/write (recommended ≥ 80 MB/s)	

		Format: FAT32 or exFAT	
04	5V Buzzer (Active)	Voltage: 5V DC Control: ON/OFF via digital signal Sound: Constant tone when powered Use: Simple sound alerts in projects	
05	Red LED (5mm)	Color: Red Forward Voltage: ~2V Current: ~20mA Use: Indicators, status lights in circuits	
06	Green LED (5mm)	Color: Green Forward Voltage: ~2.1V Current: ~20mA Use: Indicators, visual signals in circuits	
07	220Ω Resistors	Resistance: 220 ohms (Ω) Tolerance: $\pm 5\%$ (typically) Power Rating: 0.25W or 0.5W Use: Current limiting for LEDs, basic circuit protection	

08	GPIO Header Wires	Use: Connect Raspberry Pi GPIO pins to sensors, modules, breadboards Pitch: 2.54 mm standard Length: Typically 10–30 cm	
09	Power Bank (10,000 mAh)	Capacity: 10,000 mAh Output: 5V, 2A (ideal for Raspberry Pi) Ports: Typically, 1–2 USB outputs Use: Portable power for Raspberry Pi or mobile devices Feature: Rechargeable, overcharge protection	
10	Micro-USB Cable	Connector Type: USB-A to Micro-USB Use: Power and data transfer Compatibility: Raspberry Pi 3B+ and other micro-USB devices Current Rating: Recommended 2A or higher for stable Pi power Length: Typically 0.5–1 meter	
11	Mini Breadboard / PCB	Mini Breadboard: Type: Solderless	

		prototyping board • Size: ~170 tie-points • Use: Temporary circuit building	
12	Case/Enclosure (Custom)	Use: Protects and secures Raspberry Pi & components Type: 3D-printed, acrylic, or DIY mounts Features: Ventilation, GPIO access, camera slots	

Design of Main Parts of AI Based PPE Detection System

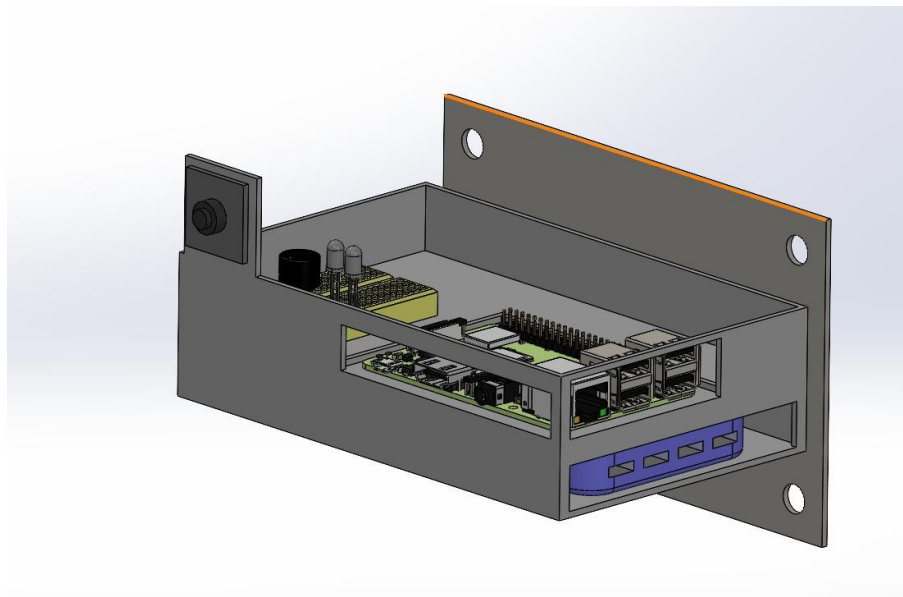


Figure 3 Isometric View of AI Based PPE Detection System

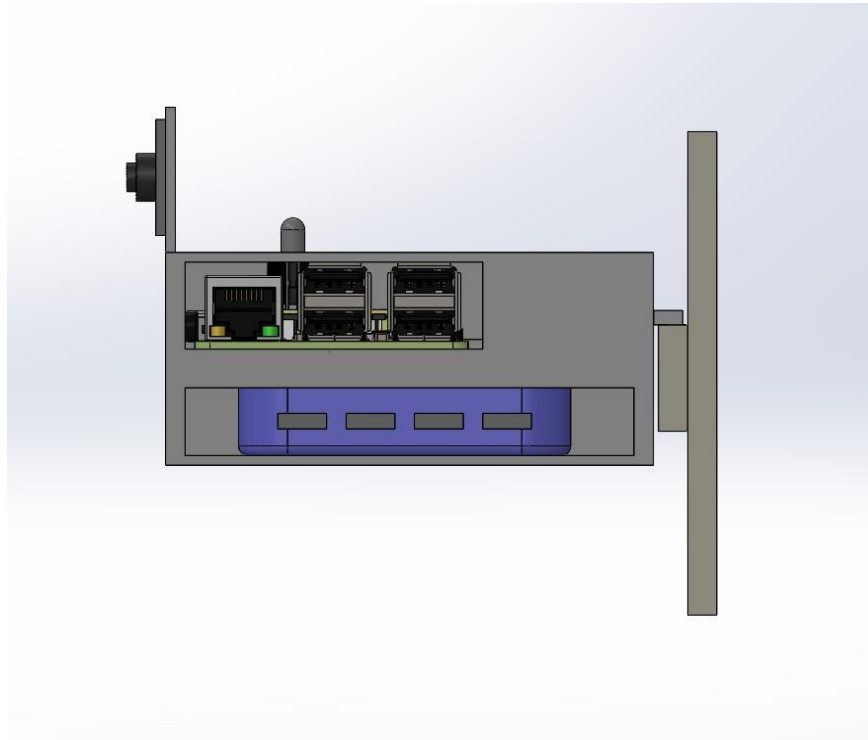


Figure 4 Side View

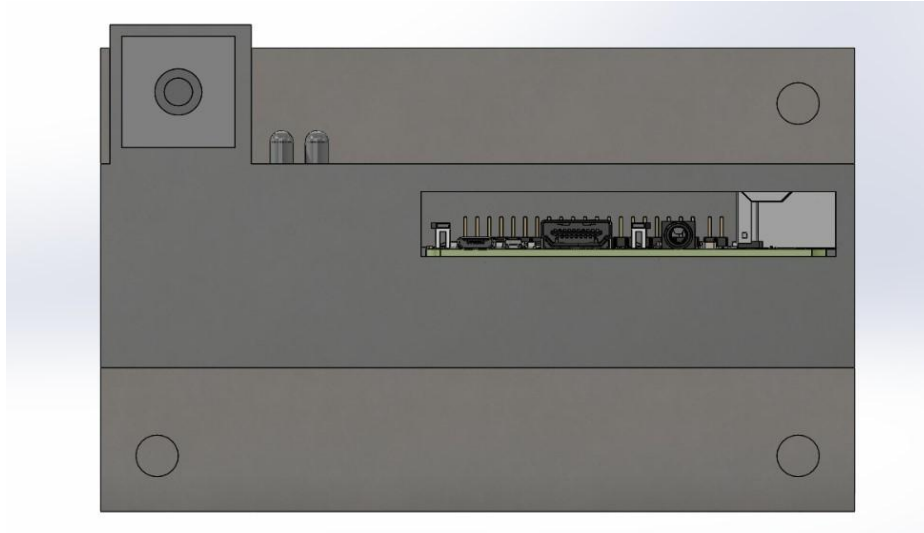


Figure 5 Front View

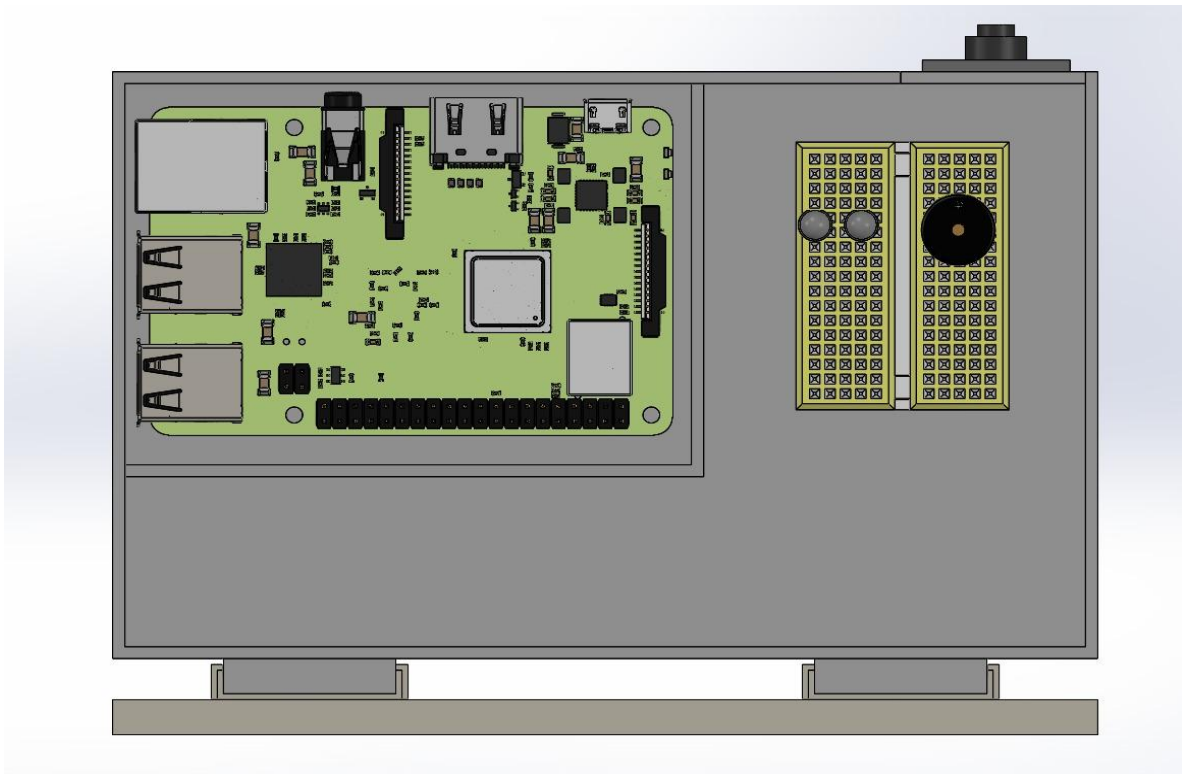


Figure 6 Top view

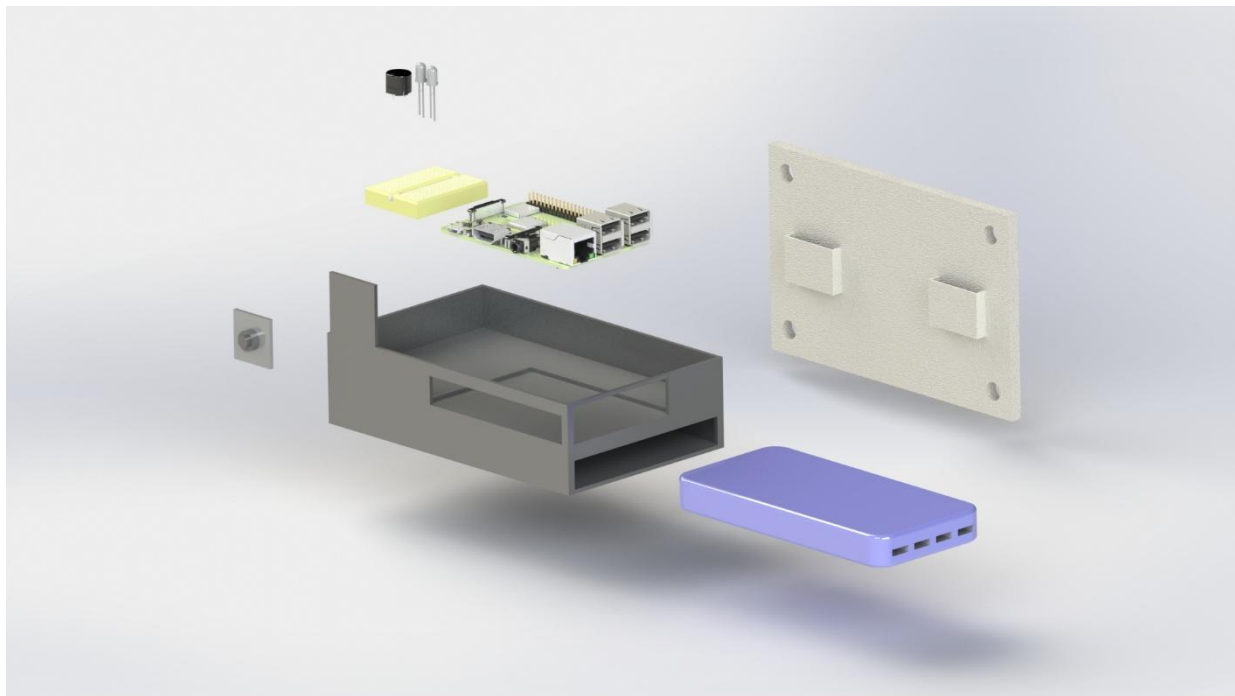


Figure 7 Exploded View

Environmental Impact

Positive Impacts:

- **Reduction of Occupational Hazards:** By ensuring consistent compliance with PPE requirements, the system helps prevent workplace injuries. This indirectly reduces the demand for medical treatment, thereby lowering associated medical waste and resource consumption.
- **Support for Sustainable Operations:** The implementation of an automated PPE compliance system encourages industries to uphold rigorous safety standards, aligning with broader goals for sustainable and responsible industrial practices.
- **Potential for Low Energy Consumption:** If designed with energy-efficient components and optimized operating modes, the system can maintain a minimal environmental footprint during its operational life cycle.

Potential Adverse Impacts:

- **Generation of Electronic Waste:** Widespread deployment over time may contribute to electronic waste due to the use of microcontrollers, sensors, and other electronic components.
- **Battery Disposal Concerns:** The inclusion of rechargeable or replaceable batteries introduces the risk of improper disposal, which can lead to soil and water contamination.
- **Cumulative Energy Use:** If not properly optimized, the collective power consumption of multiple units may become significant, especially in large-scale industrial applications.

Mitigation Measures:

To address these concerns, the system should be designed for durability and ease of recycling. Rechargeable batteries and modular components that facilitate repair and responsible end-of-life disposal are recommended. Incorporating energy-saving modes and low-power hardware will further reduce the environmental burden.

Social Impact

Positive Impacts:

- **Enhanced Worker Safety:** The primary social benefit is the significant reduction of workplace injuries and fatalities by ensuring that personnel adhere to PPE guidelines.
- **Increased Safety Awareness:** The system fosters a culture of accountability and heightened awareness regarding workplace safety, benefiting both workers and management.
- **Improved Productivity and Morale:** A safer working environment leads to fewer disruptions, greater operational continuity, and higher worker morale, as employees feel more secure and valued.
- **Employment Opportunities:** The introduction of such a system may generate new employment opportunities related to its installation, operation, and maintenance.

Potential Adverse Impacts:

- **Privacy and Autonomy Concerns:** Continuous monitoring may raise concerns among workers regarding personal privacy and perceived surveillance.
- **Resistance to Adoption:** There may be initial resistance from employees or labor unions who perceive the system as intrusive or unnecessary.
- **Financial Constraints:** Small and medium-sized enterprises may encounter challenges in bearing the initial costs of implementation.

Mitigation Measures:

To minimize these challenges, it is advisable to implement clear, transparent policies regarding data use and monitoring scope. Engaging workers during the planning and deployment phases, offering training, and providing opportunities for feedback will help build trust and acceptance. Financial support or phased rollouts may alleviate cost-related barriers for smaller organizations.

Discussion:

The lab on Functional Structure and Design & Specifications focused on analyzing the main components and their interconnections within a system or product. It emphasized how each component contributes to overall functionality and performance. During the session, we identified, labeled, and discussed the purpose of each part and examined how specifications like dimensions, materials, and power requirements affect design decisions. We also explored the relationship between form and function, considering efficiency and cost-effectiveness. This understanding is essential for optimizing real-world engineering solutions.

Conclusion:

In conclusion, the lab enhanced our understanding of how functional structure and detailed specifications guide effective product and system design. This knowledge is crucial for ensuring reliability, efficiency, and cost optimization. Industrial engineers play a vital role in this process by integrating engineering principles with management practices to improve systems, reduce waste, and maximize productivity. Their ability to design smart, efficient workflows makes them essential in today's competitive industries.